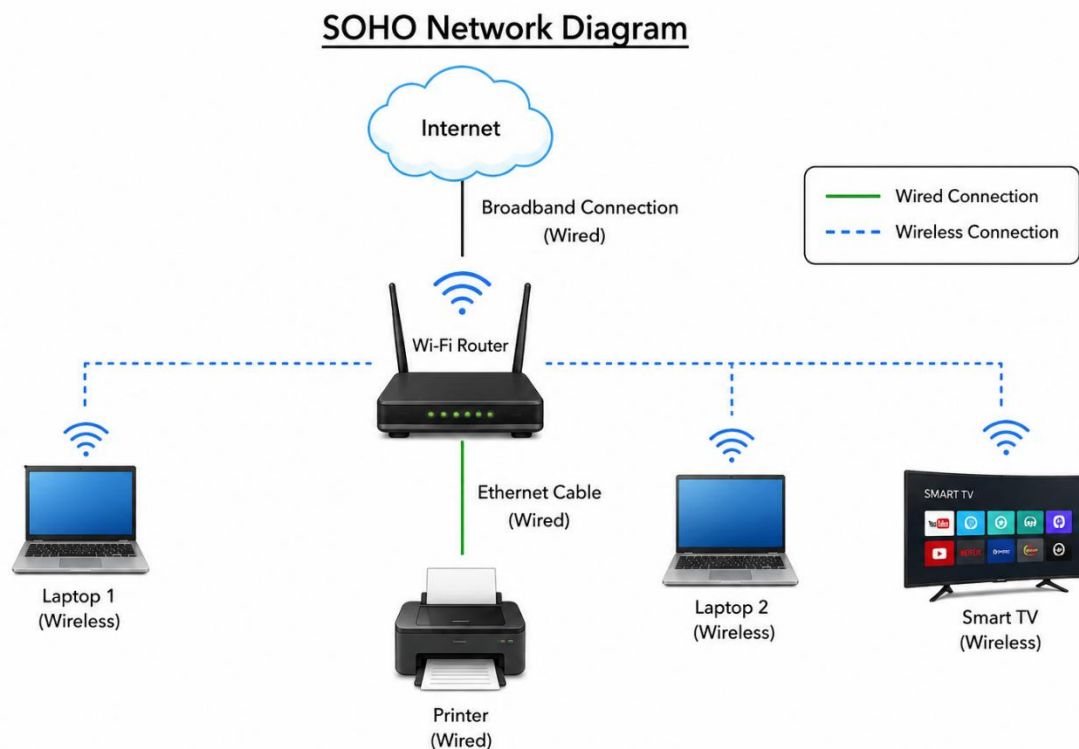


Task

- 1) Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



- 2) List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Hardware Component	Suggested Brand/Model	Purpose
Broadband Internet Connection	Jio Fiber, Airtel X-stream Fiber, BSNL FTTH	Provides internet to the office/home.
Wi-Fi Router	TP-Link Archer AX23 or TP-Link Archer AX10	Connects all devices and provides Wi-Fi and LAN ports.
Laptop 1	Dell Inspiron 15 / HP 15s / Lenovo IdeaPad Slim 3	Used by the first user.

Laptop 2	Dell Inspiron 15 / HP 15s / Lenovo IdeaPad Slim 3	Used by the second user.
Smart TV	Samsung Crystal UHD / LG UHD Smart TV / Xiaomi Smart TV	Connects to the Wi-Fi for streaming.
Printer	HP Smart Tank 585 (Wi-Fi supported) or Epson EcoTank L3211	Prints documents. HP Smart Tank supports wireless printing.
Ethernet (Cat6) Cable	Sounce Cat6 or Portronics NetX Ethernet Cable	Connects the router to the broadband modem or printer (if using a wired connection).
Power Adapter	Comes with each device	Powers the router, printer, and TV.

- 3) Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Step 1: Connect the Hardware

1. Connect the broadband internet cable (from your ISP modem/ONT) to the WAN/Internet port of the Wi-Fi router.
2. Connect your laptop to the router using a LAN (Ethernet) cable or the default Wi-Fi network.
3. Turn on the router and wait until the power and internet lights become stable.

Step 2: Log In to the Router

1. Open a web browser (Chrome, Edge, or Firefox).
2. Type the router's default IP address in the address bar. Common addresses are:
 - o 192.168.0.1
 - o 192.168.1.1
3. Press Enter.
4. Log in using the default username and password.

Step 3: Change the Default Admin Password

1. Go to Administration, System Tools, or Management.
2. Find the Change Password or Admin Password option.
3. Enter the current password.
4. Create a new strong password using letters, numbers, and symbols.

5. Save the changes.

Example:

- Username: admin
- New Password: SOHO@2026Secure

Step 4: Set a Unique Wi-Fi Name (SSID)

1. Go to Wireless Settings.
2. Locate the SSID (Wi-Fi Name) field.
3. Replace the default Wi-Fi name with a unique one.

Example SSID: Om_Home_Office_WiFi

4. Click Save or Apply.

Step 5: Enable WPA2 Security

1. In the Wireless Security section, choose:
 - o Security Mode: WPA2-PSK (AES)
2. Create a strong Wi-Fi password.

Example Wi-Fi Password: Office@12345

3. Save the settings.

Step 6: Restart the Router

1. If prompted, restart the router.
2. Wait for it to reboot completely.

Step 7: Connect Your Devices

Connect the following devices to the new Wi-Fi network:

- Laptop 1
- Laptop 2
- Smart TV

If the printer is connected by Ethernet, plug it into one of the router's LAN ports. If it supports Wi-Fi, connect it to the same Wi-Fi network.

Step 8: Test the Network

1. Open a web browser on each laptop and check if the internet is working.
2. Verify that the Smart TV can access online apps.

3. Print a test page from the printer to confirm it is connected.

- 4) Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC Address (Example)	IP Assignment	IP Address
Wi-Fi Router (Gateway)	00:1A:2B:3C:4D:01	Static	192.168.1.1
Laptop 1	A4:5E:60:12:34:56	DHCP	192.168.1.100
Laptop 2	B8:27:EB:65:43:21	DHCP	192.168.1.101
Smart TV	5C:CF:7F:89:AB:CD	Static	192.168.1.10
Printer	3C:52:82:EF:12:34	Static	192.168.1.20

Network Settings

- **Network Address:** 192.168.1.0/24
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1
- **DNS Server:** 8.8.8.8 (Google DNS) or your ISP's DNS
- **DHCP Range:** 192.168.1.100 – 192.168.1.200

Explanation

- The **Wi-Fi router** acts as the **default gateway** for all devices.
- **Laptop 1** and **Laptop 2** receive their IP addresses automatically from the router using **DHCP**.
- The **Smart TV** is assigned a **static IP address (192.168.1.10)** so it always keeps the same address on the network.
- The **Printer** is assigned a **static IP address (192.168.1.20)**, making it easy for all users to find and print to it without the IP address changing.

- 5) Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible Cause	How to Diagnose	How to Fix
1. Wi-Fi is turned off or not connected	Check if the Wi-Fi icon shows "Disconnected" or if Airplane Mode is enabled.	Turn on Wi-Fi, disable Airplane Mode, and reconnect to the correct Wi-Fi network (SSID). Enter the correct Wi-Fi password if required.
2. Laptop did not receive an IP address	Open Command Prompt and type ipconfig . If the IP address starts with 169.254.x.x , the laptop did not get an IP address from the router.	Restart the router, then type ipconfig /release followed by ipconfig /renew to request a new IP address.
3. Router or Internet connection is down	Check whether other devices (such as the second laptop or Smart TV) can access the internet. Also check if the router's Internet light is on.	Restart the router and modem. If no devices have internet, contact the Internet Service Provider (ISP).

Additional Troubleshooting Steps

- Restart the laptop and the Wi-Fi router.
- Make sure the laptop is connected to the correct Wi-Fi network.
- Check if the Wi-Fi password is correct.
- Run the Windows **Network Troubleshooter**.
- Open Command Prompt and test the connection:

○ `ping 192.168.1.1` → Tests the connection to the router.
○ `ping 8.8.8.8` → Tests whether the internet is reachable.
○ `ping google.com` → Checks whether DNS is working.